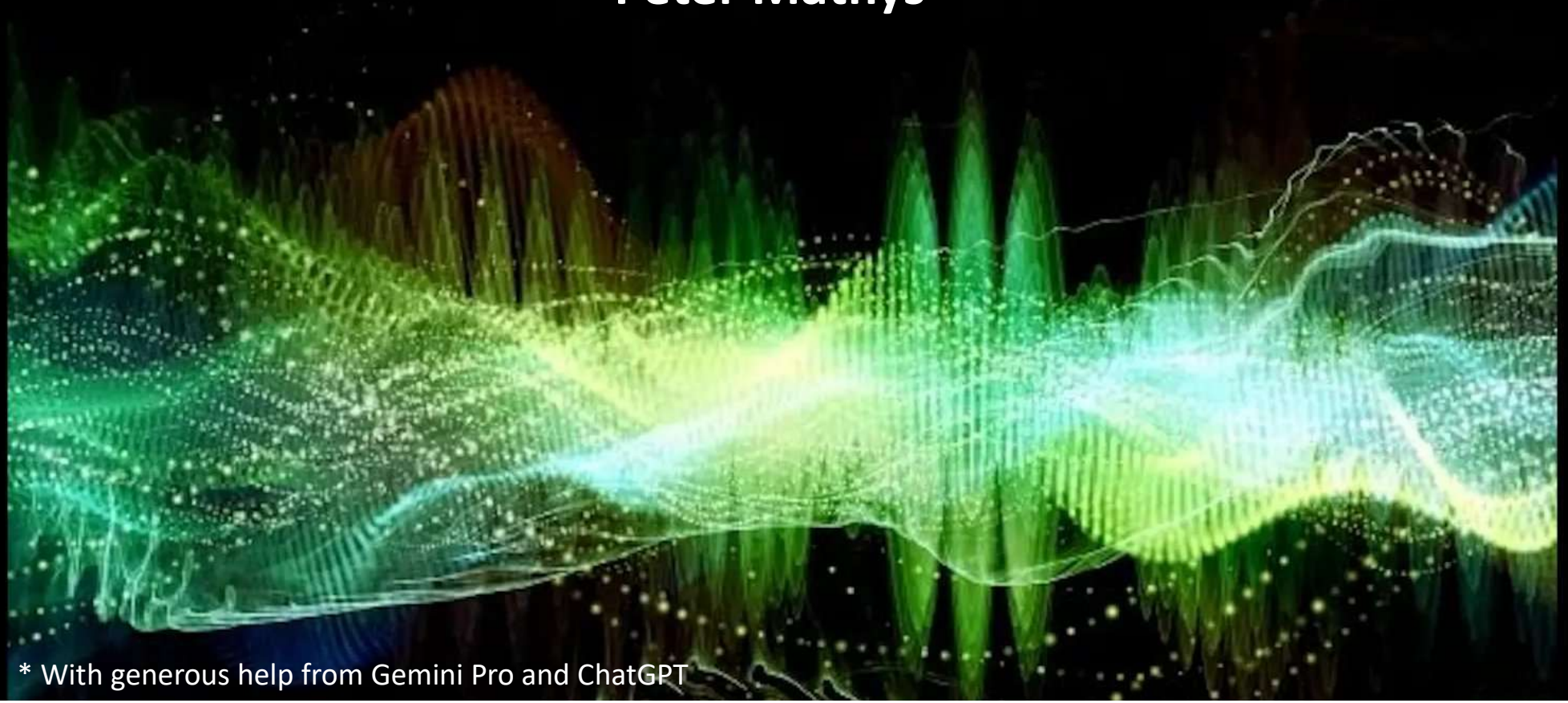
ECEN 4/5632 Theory and Application of Digital Filters
Lecture 12, 10-13-26, 10-15-26
Peter Mathys*



* With generous help from Gemini Pro and ChatGPT

WEEK 6 ● LECTURE 12

The STFT and Spectrograms

Local spectra for non-stationary signals and 2-D ML features

Oppenheim & Schafer • DFT / FFT / STFT | Raschka • CNN architecture and feature pipelines

Learning objectives

ROADMAP

0:00–0:03

- Explain why a single global spectrum can fail for non-stationary signals.
- Interpret the STFT as a sequence of windowed local Fourier transforms.
- Predict the time–frequency resolution trade-off from window duration.
- Read and construct a spectrogram.
- Format spectrograms as structured tensors for PyTorch CNNs.

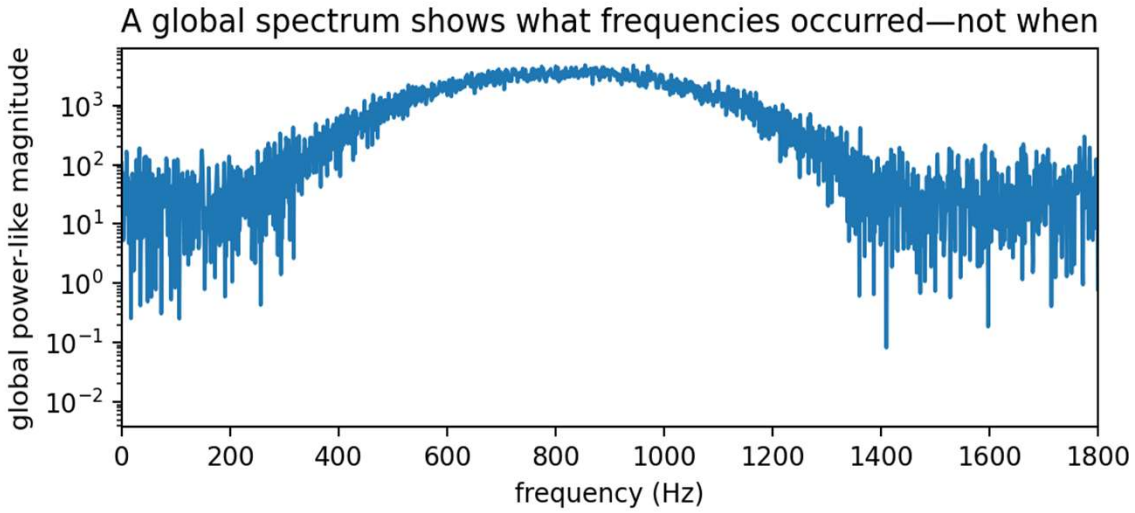
Core question

How do we preserve both when a frequency occurs and what that frequency is?

Why a global spectrum is not enough

NON-STATIONARY SIGNALS

0:03–0:10



What it tells us

Many frequencies occurred during the record.

What it loses

It does not tell us that the dominant frequency swept upward with time.

Speech, radar, machinery faults, ECG/EEG, and transient events are often non-stationary.

Localize before transforming

STFT

0:10–0:15

Take a short window around a time location, then compute a Fourier transform of that local segment.

$$X(r,\omega) = \Sigma_m\, x[m]\, w[m-rR]\, e^{(-j\omega m)}$$

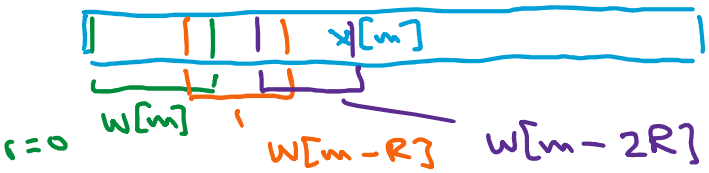
$$X(r,\omega) = \sum_m x[m]\, w[m-rR] \cdot e^{-j\omega m}$$

DTFT of blocks of x[m] multiplied by window w

- r = frame index
- R = hop size between frames
- w[n] = analysis window
- Each frame produces one local spectrum.

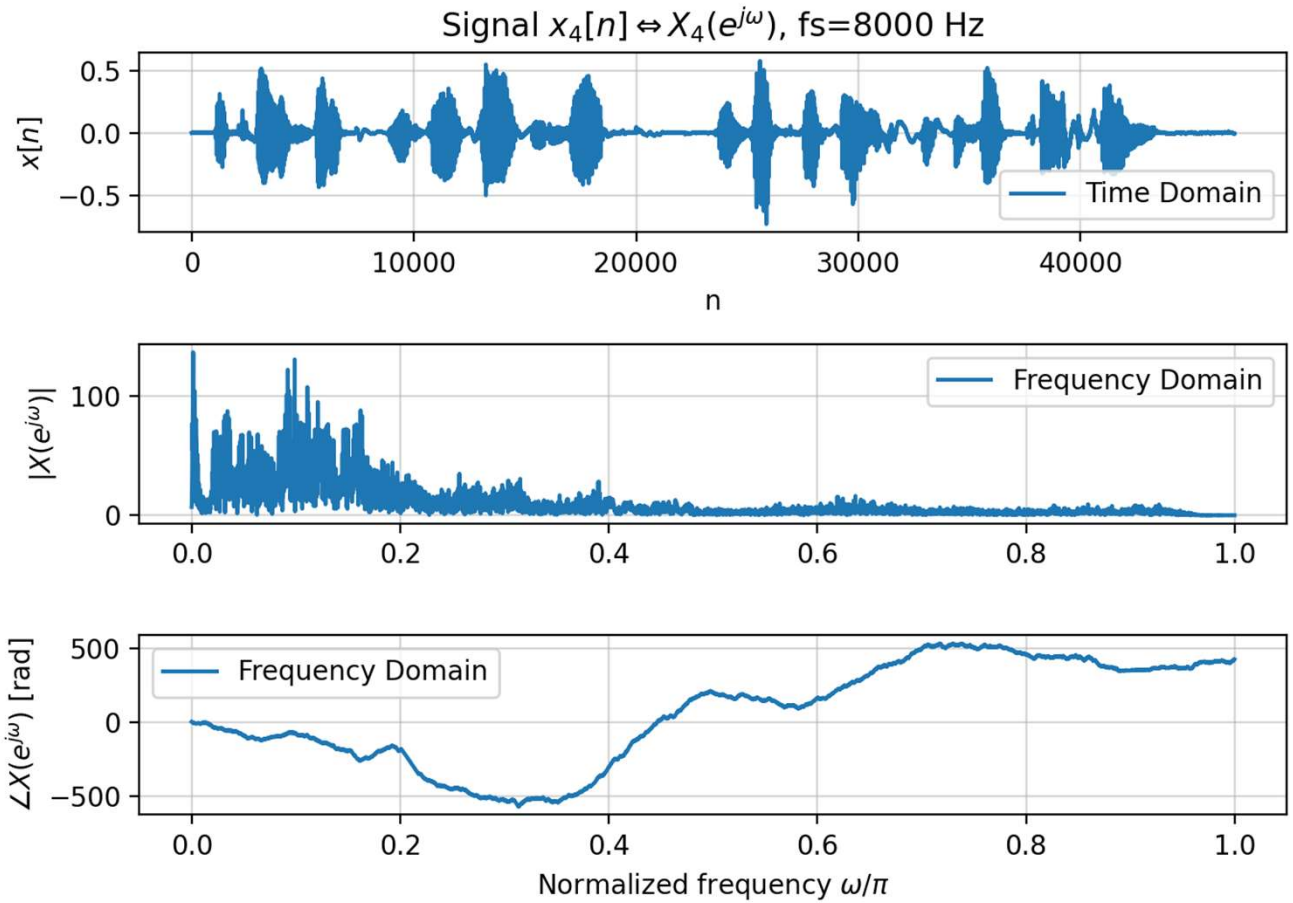
Equivalent notation

Texts place the window as w[rR−m] or w[m−rR]. For real symmetric windows, the practical idea is the same: select a neighborhood around the frame time.



R : hop length

Example: Speech Signal

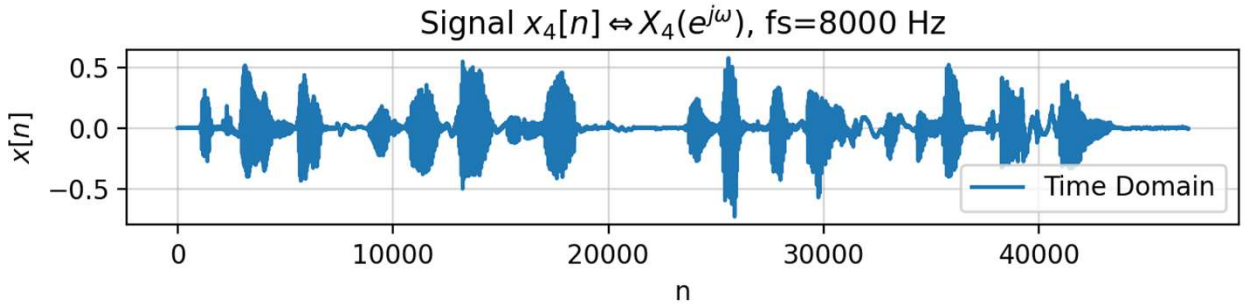


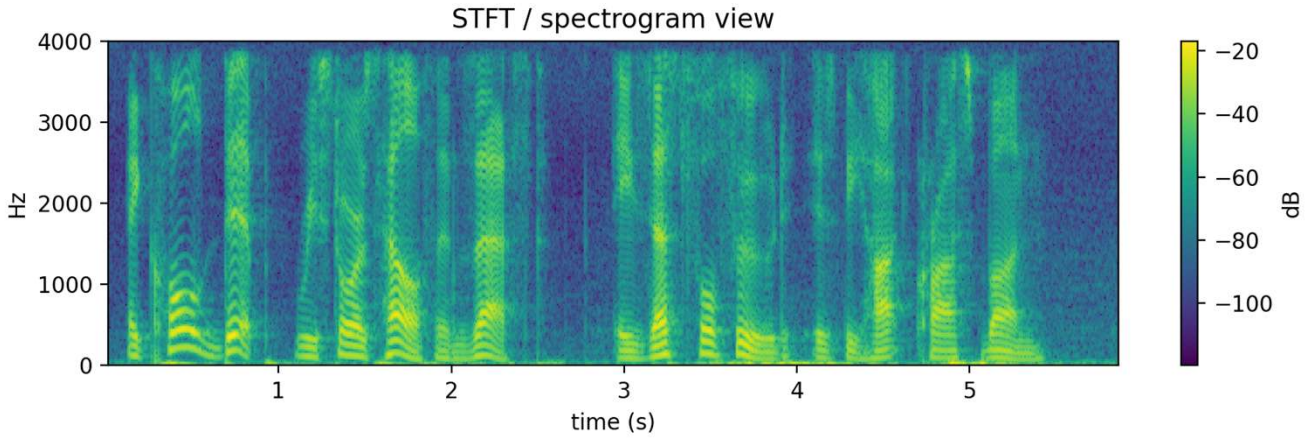
Looking only at time domain, frequency content is not visible.

Looking at magnitude in the frequency domain, time domain information is not visible.

Looking at the phase in the frequency domain does not reveal any interesting information.

Example: Speech Signal





"A cold dip restores health and zest, a zestful food is the hot cross bun"

A spectrogram visualizes both time and frequency by dividing up an audio signal into overlapping blocks and taking Short Time Fourier Transforms (STFTs) of each block that are then displayed as two dimensional images.

From frames to a time–frequency matrix

STFT

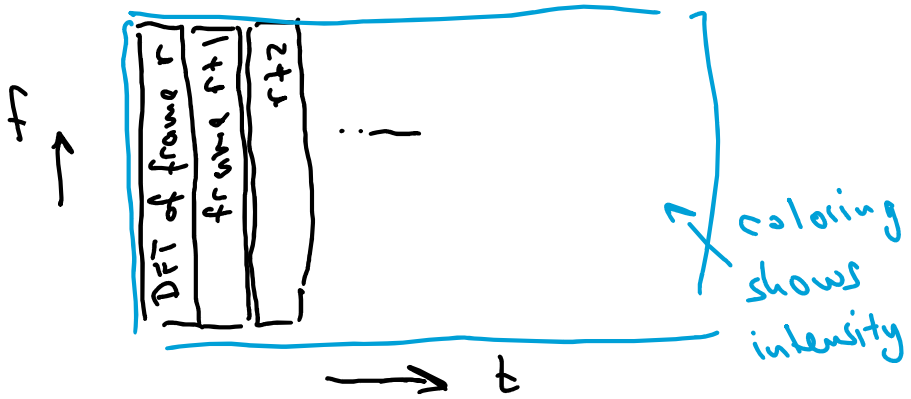
0:15–0:22

- Frame the signal into overlapping segments.
- Multiply each segment by a window.
- Take an FFT along the sample dimension.
- Stack the spectra as columns (or rows).

PyTorch viewpoint

That matrix is already a structured tensor. Add batch and channel dimensions and it looks like an image input to a CNN.

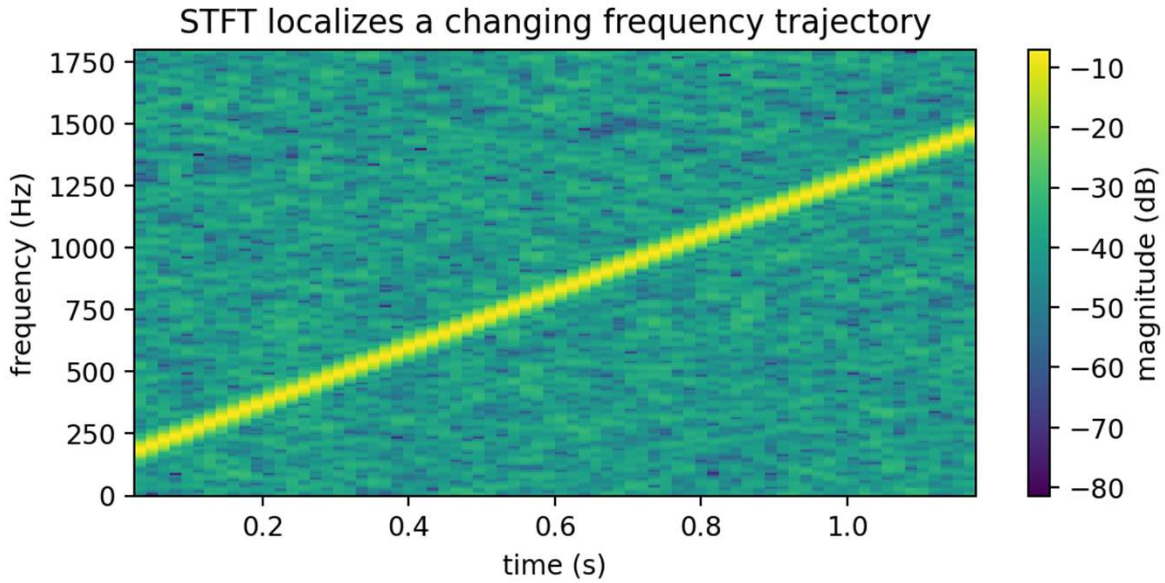
STFT output shape ≈ frequency bins × time frames



The STFT reveals the chirp trajectory

STFT

0:22–0:28



Read it

Horizontal position = time.
Vertical position = frequency.
Brightness/color = local magnitude or power.

The upward ridge is the time-varying instantaneous frequency.

Window length sets the central trade-off

RESOLUTION

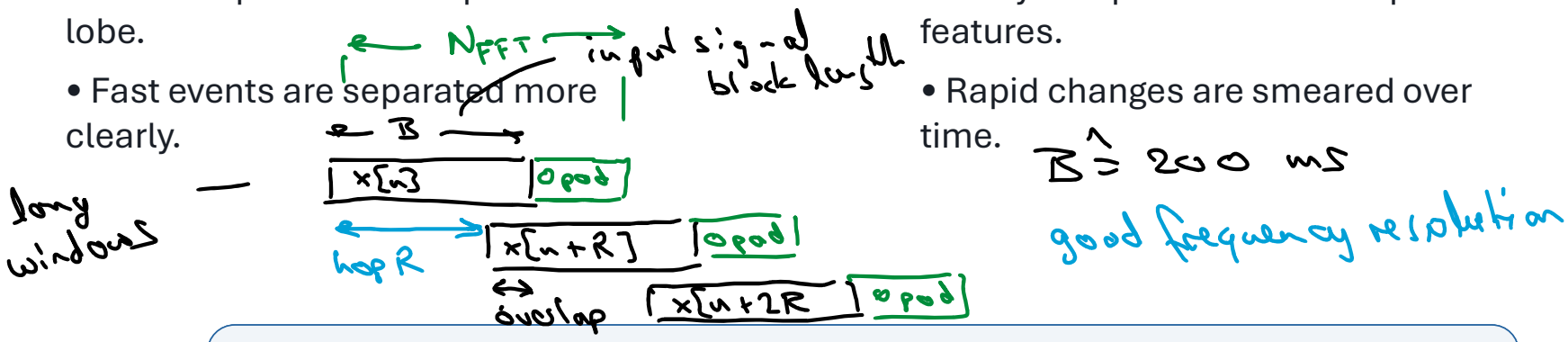
0:28–0:34

Short window

- Good time localization.
- Few samples → broad spectral main lobe.
- Fast events are separated more clearly.

Long window

- Good frequency localization.
- Many samples → narrower spectral features.
- Rapid changes are smeared over time.



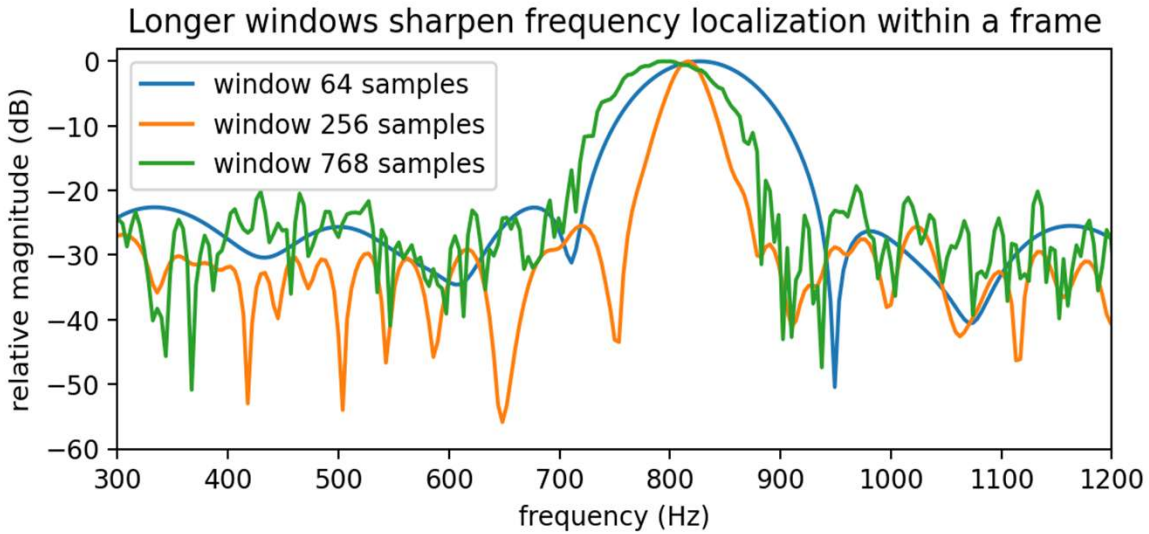
No free lunch

You cannot make a window arbitrarily narrow in time and simultaneously arbitrarily narrow in frequency.

Frequency view of the window trade-off

RESOLUTION

0:34–0:39



Interpretation

At one time frame, longer windows produce sharper frequency localization. But they average over a longer interval of the chirp.

Resolution is not the same as FFT display density.

Hop size and overlap

STFT

0:39–0:43

Hop R = number of new samples between adjacent frames

- Small R → many overlapping frames and a dense time grid.
- Large R → fewer frames, less computation, coarser time sampling.
- Overlap does not undo the fundamental window resolution trade-off.
- For reconstruction, window/hop choices must also satisfy overlap-add conditions.

Common choice

For analysis with Hann windows, 50–75% overlap is common, but the application should set the hop.

Spectrogram = squared STFT magnitude

SPECTROGRAM

0:43–0:49

$$\text{Spectrogram}(r,\omega) = |X(r,\omega)|^2$$

mag squared

- Discard phase and visualize local energy/power-like content.
- Often displayed in decibels: $10 \log_{10}(\text{power} + \varepsilon)$.
- Color scaling strongly affects what patterns are visible.
- One-sided display is common for real-valued signals.

Important

A spectrogram is not just a pretty picture—it is a numerical matrix whose axes and normalization matter.

Power Spectral Density PSD

For random process $x[n]$ if it is wide-sense stationary (WSS)

Autocorrelation function of random process $x[n]$

$$R_x[m] = E[x[n] \cdot x^*[n-m]] = E[x[k+m]\, x^*[k]]$$

$$\Updownarrow \text{ DTFT}$$

$$S_x(\omega) = \sum_{m=-\infty}^{\infty} R_x[m]\, e^{-j\omega m}$$

Wiener-Khintchine theorem

$\uparrow$

PSD of WSS process $x[n]$

Spectrogram $|X(r,\omega)|^2$

$$X(r,\omega) = \sum_m x[m]\, w[m-rR]\, e^{-j\omega m}$$

$n \leftarrow m - rR$

$n + rR \leftarrow m$

$$\Downarrow$$

$$X(r,\omega) = \sum_n x[n+rR]\, w[n]\, e^{-j\omega n}$$

frame size

assume window of size B

$$\Rightarrow X(r,\omega) = \sum_{n=0}^{B-1} x_r[n]\, w[n]\, e^{-j\omega n}$$

$x_r[n] = x[n+rR]$

now we can use DFT/FFT of length N, $\omega \to \frac{2\pi k}{N}$

$$\Rightarrow X_r[k] = \sum_{n=0}^{B-1} x_r[n]\, e^{-j2\pi kn/N}$$

$N \geq B$

normalization $P_r[k] = \dfrac{|X_r[k]|^2}{\sum_{n=0}^{B-1} |w[n]|^2}$

modified periodogram

Periodogram

normalization $P_r[k] = \frac{|X_r[k]|^2}{\sum_{n=0}^{N-1} |w[n]|^2}$

modified periodogram

Welch's method averages over modified periodograms to obtain better estimates of PSD for noisy signals

standard periodogram

$$P_{per}[k] = \frac{1}{N} \left| \sum_{n=0}^{N-1} x[n] \, e^{-j 2\pi k n / N} \right|^2$$

↑ for whole $x[n]$ (finite record of length N)

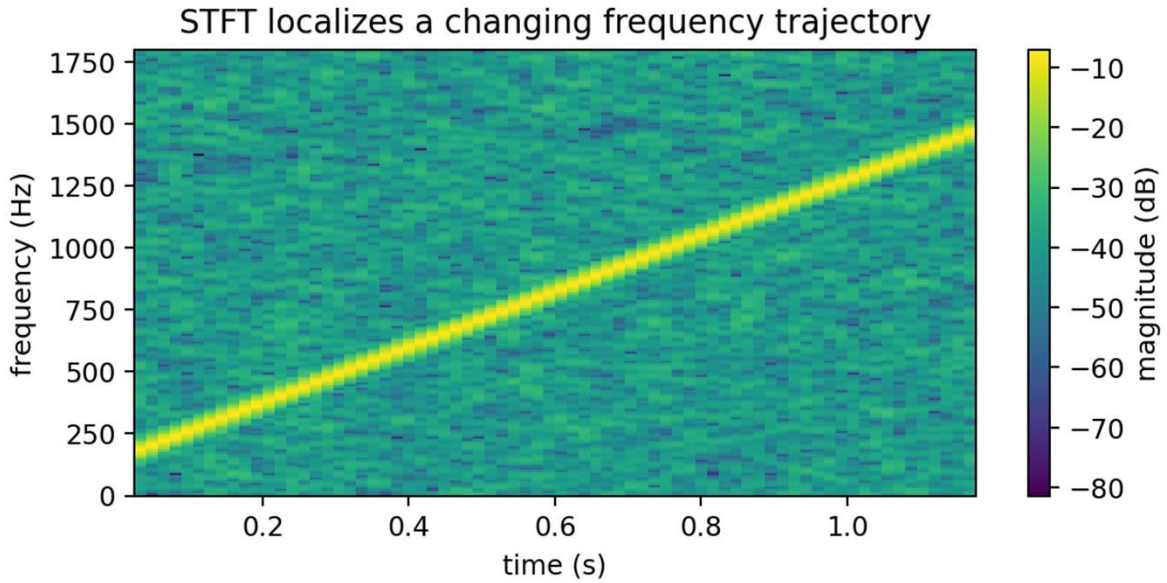
estimate of PSD of $x[n]$

Parseval's Identity: $\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2$

How to read a spectrogram

SPECTROGRAM

0:49–0:55



- Horizontal ridge → sustained tone.
- Vertical stripe → broadband transient.
- Sloped ridge → chirp or changing pitch.
- Harmonic stacks → periodic voiced or rotating behavior.
- Diffuse background → broadband noise.

Ask students

What time-domain event would create a narrow vertical stripe across many frequencies?

Concept check: two close tones that switch

CONCEPT CHECK

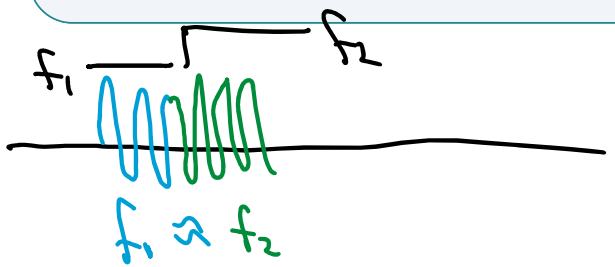
0:55–0:58

A signal contains 500 Hz for the first 50 ms and 550 Hz for the next 50 ms.

Would you begin with a very short or very long STFT window?

Good answer

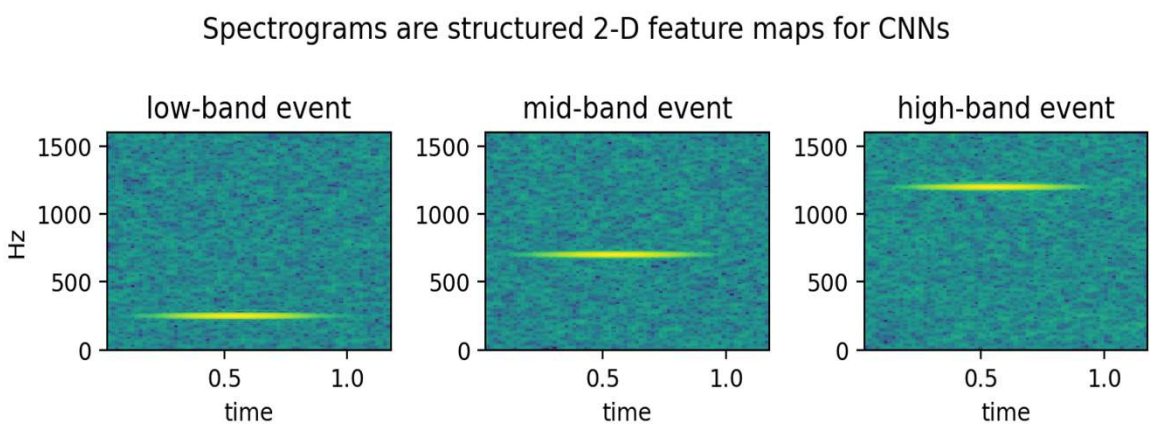
Neither extreme. The window must be long enough to distinguish 50-Hz-separated tones but short enough not to smear the 50-ms transition. The problem is inherently a time–frequency trade-off.



Spectrograms as CNN inputs

ML BRIDGE

0:58–1:04



- Frequency bins behave like a vertical spatial axis.
- Time frames behave like a horizontal spatial axis.
- Local 2-D filters can learn ridges, transients, harmonic patterns, and modulations.
- Typical tensor shape: [batch, channel, frequency, time].

Example: [32, 1, 257, 101]

Magnitude scaling and normalization before ML

ML BRIDGE

1:04–1:08

- Use magnitude or power consistently.
- Log compression reduces extreme dynamic range and often makes weak structure more visible.
- Normalize using training-set statistics, not statistics from the test set.
- Keep sample rate, STFT parameters, and frequency range fixed across examples.
- Avoid data leakage when computing normalization or augmentations.

Pipeline reproducibility

The STFT is part of the model input definition. Changing window length or hop changes the feature space just like changing a network layer.

PyTorch: from waveform to image tensor

ML BRIDGE

1:08–1:11



waveform [B,N] → STFT → magnitude → log → [B,1,F,T] → Conv2d




```
Z = torch.stft(x, n_fft=512, hop_length=128, window=win, return_complex=True)
```

```
S = torch.log1p(torch.abs(Z)**2).unsqueeze(1)
```

Why Conv2d?

The model can exploit local patterns jointly across neighboring times and neighboring frequencies.

When not to treat the spectrogram as a generic image

ML BRIDGE

1:11–1:13

- Time and frequency axes are not interchangeable.
- A one-bin translation in frequency may have a different physical meaning at different scales.
- Image augmentations such as arbitrary rotation or vertical flipping are usually physically nonsensical.
- DSP-informed augmentations should preserve plausible signal behavior.

Good ML habit

Use domain knowledge to decide which invariances the model should learn rather than importing image tricks blindly.

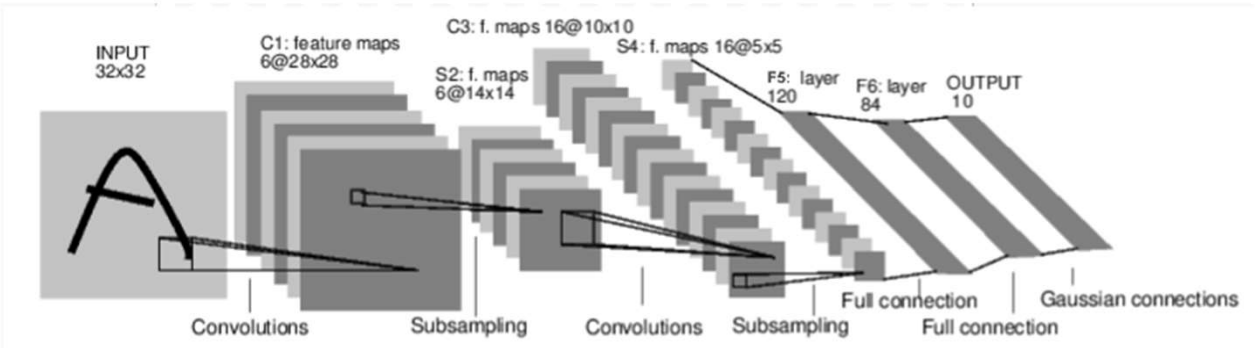
6. PyTorch spectrogram tensor and tiny CNN

```
try:
    import torch
    from torch import nn
    xt=torch.tensor(x,dtype=torch.float32).unsqueeze(0)
    win=torch.hann_window(256)
    Zt=torch.stft(xt,n_fft=512,hop_length=64,win_length=256,window=win,return_complex=True)
    S=torch.log1p(torch.abs(Zt)**2).unsqueeze(1)
    net=nn.Sequential(nn.Conv2d(1,8,3,padding=1),nn.ReLU(),nn.MaxPool2d(2),
                      nn.Conv2d(8,16,3,padding=1),nn.ReLU(),nn.AdaptiveAvgPool2d((1,1)),
                      nn.Flatten(),nn.Linear(16,3))
    logits=net(S)
    print('spectrogram tensor:',S.shape)
    print('CNN logits:',logits.shape)
except Exception as e:
    print('PyTorch demo skipped:',e)
```

```
spectrogram tensor: torch.Size([1, 1, 257, 76])
CNN logits: torch.Size([1, 3])
```

```
x.shape
```

```
(4800,)
```



Lecture 12 wrap-up

WRAP-UP

1:13–1:15

- A global transform loses when frequency content occurs.
- The STFT computes local windowed spectra over time.
- Window duration trades time resolution against frequency resolution.
- The spectrogram is $|\mathrm{STFT}|^2$ and is a structured time–frequency matrix.
- Spectrograms create a direct bridge from DSP to 2-D CNNs.

Exit ticket

A transient lasts 8 ms. Why might a 200-ms STFT window be a poor choice even if it gives excellent frequency resolution?